```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title> The Material Part 2 Learn HTML 5 Modern </title>
</head>
<body>
    <!-- Continue Material Part 1 - Learn HTML 5 Modern -->


            <!-- Chapter 6 - Table-->
    <h2> Chapter 6 : Tables </h2>


    <p> The HTML <b>(table)</b> element allowws web authors to display tabular data (such as text, images, links, other tables, etc.) in a two dimensional table with rows and colums of cells.</p>


    <!-- Chapter 6.1 - Simple Table -->
    <h3>  6.1 - Simple Table </h3>


    <table>
        <tr>
            <th> Heading 1/Colum 1 </th>
            <th> Heading 2/Colum 2 </th>
        </tr>
        <tr>
            <td> Row 1 Data Column 1 </td>
            <td> Row 2 Data Column 2 </td>
        </tr>
        <tr>
            <td> Row 2 Data Column 1</td>
```

```
        <td> Row 2 Data Column 2</td>
    </tr>
</table>

<p> This will render a  <b>(table)</b> consisting of three total rows <b>(tr)</b>: one row of header cell <b>(th)</b> and two rows of content cell <b>(td)</b>. <b>(th)</b> elements are <i>tabular header</i> and <b>(td)</b> elements are <i>tabular data</i>. You can put whatever you want inside a <b>(td)</b> or <b>(th)</b> </p>

<p>
    <b>Heading 1/Column 1Heading 2/Column 2</b>
    <br> <!-- Spacing line like enter -->
    Row 1 Data Column 1Row 1 Data Column 2 <br>
    Row 2 Data Column 1Row 2 Data Column 2  <br>
</p>

<!-- Chapter 6.2 - Simple Table -->
<h3>  6.2 - Spanning columns or rows </h3>
<p> Table cells can span multiple columns or rows using the colspan and rowspan attributes. These attributes can be applied to <b>(th)</b> and <b>(td)</b> elements.</p>

<table>
    <tr>
        <td> row 1 col 1 </td>
        <td> row 2 col 2 </td>
        <td> row 3 col 3 </td>
    </tr>
    <tr>
        <td colspan="3"> This second row spans all three columns </td>
    </tr>
    <tr>
        <td rowspan="2"> This Cell spans two rows </td>
```

```
        <td> row 3 col 2 </td>
        <td> row 3 col 3 </td>
      </tr>
      <tr>
        <td> row 4 col 2 </td>
        <td> row 4 col 3 </td>
      </tr>
    </table>
```

 Note that you should not design a table where bot rows and colums overlap as this is invalid HTML and the result is handled differently by different web browsers.

 rowspand = A non-negative integer that specifies the number of rows spanned by a cell. The default value of this attributes is one (1). A value of zero (0) means that the cell will extend from the current row until the last row of the table (thead), (tbody), (tfoot).

 colspan = A non-negative integer that specifies the number of columns spanned by the current cell. The default value of this attribute is one (1). A value of zero (0) means that the cell will extend from the current to the last column of the column group (colgroup) in which the cell is defined

```
    <!-- Chapter 6.3 - Column Groups -->
    <h3>  6.3 - Column Groups </h3>
```

 Sometimes you may want to apply styling to a column or group of column, or for semantic purposes, you may want to group columns together. To do this, use (colgroup) and (col) elements.

 The optional (colgroup) tag allows you to group columns together. (colgroup) elements must be child elements of a (table) and must come after any (caption)elements and before any table content (e.g., (tr), (thead), (tbody), etc).

```
    <table>
```

```
    <colgroup span="2"> </colgroup>
    <colgroup span="2"> </colgroup>
    ...
</table>
```

 The optional (Col) tag allows you to reference individual columns or a range of columns without applying a logical grouping. (col) elements are optional, but if present, they must be inside a (colgroup) element.

<!-- Spacing line like enter -->

Input:

```
<br> <!-- Spacing line like enter -->
<br> <!-- Spacing line like enter -->

<b>(table)</b>  <br>
    <b>(colgroup)</b>  <br>
        <b>(col id="MySpecialColumn" /)</b>    <br>
        <b>(col /)</b>    <br>
    <b>(/colgroup)</b>     <br>
    <colgroup>      <br>
        <b>(col class="CoolColumn")</b>    <br>
        <b>(col class="Neatcolumn" span="2" /)</b> <br>
    <b>(/colgroup)</b>     <br>
<b>(/table)</b>    <br>

<!-- Then Ouput Is: -->

<table>
    <colgroup>
```

```
        <col id="MySpecialColumn" />
        <col />
    </colgroup>
    <colgroup>
        <col class="CoolColumn">
        <col class="Neatcolumn" span="2" />
    </colgroup>
</table>

<p> The Follow CSS styles can be applied to <b>(colgroup)</b> and <b>(col)</b> elements:</p>
<ul>
    <li> Border </li>
    <li> Background </li>
    <li> Width </li>
    <li> Visibility </li>
    <li> Display (as in display : none)
        <ul>
            <li> Display : none ; will actually remove the column from the display, causing the table to render as if those cells don't exist </li>
        </ul>
    </li>
</ul>

<p> input This code: </p>

<b>(ul)</b>
    <b>(li)</b> Border <b>(/li)</b>  <br>
    <b>(li)</b> Background <b>(/li)</b> <br>
    <b>(li)</b> Width <b>(/li)</b>  <br>
    <b>(li)</b> Visibility <b>(/li)</b> <br>
```

(li) Display (as in display : none)

(ul)

(li) Display : none ; will actually remove the column from the display, causing the table to render as if those cells don't exist (/li)

(/ul)

(/li)

(/ul)

 For more information, see HTML5 Tabular Data

<!-- Chapter 6.4 - Table with thead, tbody, tfoo, and caption -->

<h3> 6.4 - Table with thead, tbody, tfoo, and caption </h3>

 HTML also provides the tables with the (htead), (tbody), (tfoot), and (caption) element. These additional elements are useful for adding semantic value to your tables and for providing a place for separate CSS styling.

 When Printing out a table that doesn't fit onto one (paper), page, most browsers repeat the contents of (thead) on every page.

 There's a specific order that must be adhered to, and we should be aware that not every element falls into place as one would expert. The Following example demonstrates how our 4 elements should be placed.

 <!-- Spacing line like enter -->

<table>

<caption> Table Title </caption>

<thead> <!-- ========== | thead is after caption | -->

<tr>

<th> Header Content 1 </th>

<th> Header Content 2 </th>

```
            </tr>
        </thead>
        <tbody> <!-- ==================== | tbody is after thead | -->
            <tr>
                <td> Body Contents 1 </td>
                <td> Body Contents 2 </td>
            </tr>
        </tbody>
        <tfoot> <!--| tfoot can be placed before or after tbody, but not in a group of tbody. | -->

        <!-- | Regardless where tfoot is in markupm it is rendered at the buttom.-->
            <tr>
                <td> Footer Content 1 </td>
                <td> Footer Content 2 </td>
            </tr>
        </tfoot>
    </table>

    <p> The following examples's results are demonstrated twice--the first table lacks any styles, the second table has a few CSS properties applied: background-color, color, and border*. The Styles are provides as a visual guide and is not an essential aspect of the topic at hand</p>

    <!-- Emmet HTML Code is table>(thead>tr>th*2+tr*6>td*2)-->

    <table>
        <thead>
            <tr>
                <th> <b>Element</b> </th>
                <th> <b>Styles Applies</b> </th>
            </tr>
```

```
        <tr>
            <td> <b>(caption)</b></td>
            <td>Yellow text on black background</td>
        </tr>
        <tr>
            <td> <b>(thead)</b> </td>
            <td> Bold text on purple background </td>
        </tr>
        <tr>
            <td> <b>(tbody)</b> </td>
            <td> Text on blue background </td>
        </tr>
        <tr>
            <td> <b>(tfoot)</b> </td>
            <td> Text on green background </td>
        </tr>
        <tr>
            <td> <b>(th)</b> </td>
            <td> Orange Borders </td>
        </tr>
        <tr>
            <td> <b>(td)</b> </td>
            <td> Red Borders </td>
        </tr>
    </thead>
</table>

<!-- Chapter 6.5 - Heading Scope -->
<h3>  6.5 - Heading Scope </h3>
```

 th elements are very commonly used to indicate headings for table rows and columns, like so:

```
<!-- emmet code : table>(thead>tr>td+th*2)>(tbody>tr*2>th+td*2)-->

<table>
    <thead>
        <tr>
            <td></td>
            <th> Column Heading 1 </th>
            <th> Column Heading 2 </th>
        </tr>
    </thead>
    <tbody>
        <tr>
            <th> Row Heading 1 </th>
            <td></td>
            <td></td>
        </tr>
        <tr>
            <th> Row Heading 2 </th>
            <td></td>
            <td></td>
        </tr>
    </tbody>
</table>
```

 This can be improved for accessibility by the use of the scopte attribute. The above example would be amended as follows:

```
<!-- Emmet code : table>(thead>tr>td+th*2)>(tbody>tr*2>th+td*2)-->
```

```
<table>
    <thead>
        <tr>
            <td></td>
            <th scope="col"> Column Heading 1 </th>
            <th scope="col"> Column Heading 2 </th>
        </tr>
    </thead>
    <tbody>
        <tr>
            <th scope="row"> Row Heading 1 </th>
            <td></td>
            <td></td>
        </tr>
        <tr>
            <th scope="row"> Row Heading 2 </th>
            <td></td>
            <td></td>
        </tr>
    </tbody>
</table>

    <p> Scope is known as an <i>enumarated  attribute</i>, meaning that it can be have a value from a specific set of possible values.
    <br> <!-- Spacing line like enter -->
    This set includes:
    </p>
    <ul>
        <li>Col</li>
        <li>row</li>
```

```
    <li>colgroup</li>
    <li>rowgroup</li>
  </ul>

  <p> Refences: </p>

  <ul>
    <li><a href="https://developer.mozilla.org/en-US/docs/Web/HTML/Element/th#attr-scope"> https://developer.mozilla.org/en-US/docs/Web/HTML/Element/th#attr-scope </a></li>
    <li><a href="https://www.w3.org/TR/WCAG20-TECHS/H63.html"> https://www.w3.org/TR/WCAG20-TECHS/H63.html</a></li>
  </ul>

  <br> <!-- Spacing line like enter -->

        <!-- Chapter 7 - Table-->
  <h2> Chapter 7 : Comments </h2>

  <p> Simmilar to other programming, markup, and markdown language, comments in HTML provides other developers with development specific information without affecting the user interfaces. Unlike other languages however, HTML comments can be used to specify HTML elements for internet Explore only. This topic explains how to write HTML comments, and their functional applications .</p>

  <!-- Chapter 7.1 - Creating comments -->
  <h3>  7.1 - Creating comments </h3>

  <p> HTML comments can be used to leave notes to yourself or other developers about a specific point in code. They can be intiated with (!-- and concluded--), like so:</p>

  <p> <i> (!-- I'm an HTML comment! --) </i></p>
```

 They can be incorporated inline within other content:

 (h1) This Part will be displayed (!-- While this will not be displayed --). (/h1)

 They can also span multiple lines to provide more information:

 (!-- This is a multiline HTML comment. Whatever is in here will not be rendered by the browser. You can "comment out" entire sections of HTML code --)

 However, they cannot appear within another HTML tag, like this:

 (h1) (!-- testAttribute="something"--) This will not work (/h1)

 (h1) (!-- testAttribute="something"--) block would be considered a single start tag h1 with some other invalid information contained within it, followed by a single > closing bracket that does nothing

 For compability with tools that try to parse HTML as XML or SGML, the body of your comment should not contain two dashes--.

<!-- Chapter 7.2 - Commenting out whitespace between inline elements -->

<h3> 7.2 - Commenting out whitespace between inline elements </h3>

 Inline display elements, usually such as span or a, will include up to one white-space character before and after them in the document. In order to Avoid very long lines in markup (that are hard to read) and unintentional white-space (which affects formatting), the white-space can be commented out.

 (-- Use an HTML comment to nullify the newline character below: --)

(a href="#") I hope there will be no extra whitespace after this! (/a) (!--
--) (button) foo (/button)

 Then Output is :

 I hope there will be no extra whitespace after this! <!---->
<button>foo</button>

 Try it without a comment between the inline elements, and there will be one space between them. Sometimes picking up the space character is desire.

 Example code:

 (-- Use an HTML comment to nullify the newline character below: --)

 (a href="#") I hope there will be no extra whitespace after this! (/a)
(!----) (button) Foo (/button)

(!-- Without it, you can notice a small formatting difference: --)

(a href="#") I hope there will be no extra whitespace after this! (/a)

(button) Foo (/button)

 Then Output Is:

 <!-- Use an HTML comment nullify the newline character below: -->

 I hope there will be no extra whitespace after this! <!---->
<button>Foo</button>

```
<hr>

<!-- Without it, you can notice a small formatting difference: -->
<a href="#">I hope there will be no extra whitespace after this!</a>
<button>Foo</button>

<br> <!-- Spacing line like enter -->

<!-- Chapter 8 - Classes and IDs -->
<h2> Chapter 8 - Classes and IDs </h2>

<!-- Emmet HTML Code : table>(thead>tr>th*2+tr*2>td*2-->
<table>
    <thead>
        <tr>
            <th><b>Parameter</b></th>
            <th><b>Details</b></th>
            <tr>
                <td>Class</td>
                <td>Indicates the Class of the element (non-unique)</td>
            </tr>
            <tr>
                <td>id</td>
                <td>Indicates th ID of the element (unique in the same context) </td>
            </tr>
        </tr>
    </thead>
</table>

    <p> Classes and IDs make referencing HTML elements from scripts and stylesheets easier. The class attribute can be used on one or more tags and is used bby CSS for styling. IDs
```

however are inteded to refer to a single element, meaning the same ID should never be used twice. IDs are generally used with JavaScript and internal document links, and are discouraged in CSS. This topic contains helpful explanations and examples regarding proper usage of class and ID attributes in HTML.

<!-- Chapter 8.1 - Giving an element a class -->

<h3> 8.1 - Giving an element a class </h3>

Classes are identifiers for the elements that they are assigned to. use the class attribute to assign a class to an element.

(div class="class 1 class 2") (/div)

<h3>Using Classes in CSS</h3>

 Classes can be used for styling certain elements without changin all elements of that kind. For Example, these two span elements can have complatelu different styling:

Input the code is:

 <!-- Spacing line like enter -->

 <!-- Spacing line like enter -->

(span) (/span)

(span class="special") (/span)

 Classes of the same name cna be given to any number of elements on a page and they will all receive the styling associated with that class. This will always be true unless you specify the element within the CSS.

 For example, we have two elements, both with the class highlight:

.highlight {color : green;}

 However if we only want to target div's with the class highlight then we can add specificity like below:

div.hightlight { color: green;}

Nevertheless, when styling with CSSm it is generally recommended that only classes (e.g. .highlight) be used rather than elements with classes (e.g .highlight).

As with any other selector, classes can be nested:

.main .highlight {color: red;} /*Descandant combinator*/

.footer > .highlight {color: blue;} /*child combinator*/

 You can also chain the class selector to the only select elements that have a combination of several classes. For example, if this is our HTML:

 (div class="special left menu") This text will be pink (/div)

Then Output is:

 <!-- Spacing line like enter -->

 <!-- Spacing line like enter -->

 This text will be pink

 And we want to colour this specific piece of text pink, we can do the following in our CSS:

.special.left.mene{color: pink;}

<!-- Chapter 8.2 - Giving an element an ID -->

<h3> 8.2 - Giving an element an ID </h3>

 The ID attribute of an element is an identifier which must be uniqe in the whole document. its purpuse is to uniqely identify the element when linking (using an anchor), scripting or styling (wiht CSS).

 (div id="example-id") (/div)

 You should not have two elements with the same ID in the same document, event if the attributes are attached two different kinds of elements. For example, the following code incorrect:

(div id="example-id") (/div)

(span id="example-id") (/span)

 Browser will do their best to render this code, but unexpected behavior may occur when styling with CSS or adding functionality with JavaScript.

To Reference elements by their ID in CSS, prefix in the ID with #.

#example-id{color:green;}

 To jumpp to an element with an ID on a given page, append # with the element name in the URL.

 http://example.com/about#example-id

 This feature is supported in most browsers and does not required additional JavaScript or CSS to work.

<!-- Chapter 8.3 - Acceptable Values -->

<h3> 8.3 - Acceptable Values </h3>

<h4> For an ID </h4>

 Version ≥ 5

 The only restrictions on the value of an id are:

 It must be unique in the document

 It must not contain any space characters

 It must containt at least one character

 So the value can be all digits, just punctuation character, include special characters, whatever. just no whitespace.

 So these are valid:

<!-- Emmet HTML code is div[id]*9{...}-->

    (div id="container") ... (/div)
    (div id="999") ... (/div)
    (div id="#%LV-||")...(/div)
    (div id="_ _ _ _V") ... (/div)
    (div id=""⌘⌥") ... (/div)
    (div id="♥") ... (/div)
    (div id="{}") ... (/div)
    (div id="©") ... (/div)
    (div id="♤₩¤☆€~¥")... (/div)

 Then Output Is:

...
...
...
...
...
...
...
...
...

 This is Invalid

...

 This is also invalid, when included in the same document

(div id="results") ... (/div)
(div id="results") ... (/div)

 Then Output is:

<!-- emmet code is did[id="result"]*2{...}-->

...
...

 version ≤ 4.01

 An id Value begin with a letter, which can then be followed only by :

<!-- emmet code is ul>li*5-->

 letters (A-Z/a-z)
 digits (0-9)
 hyphens ("_")
 underscores ("_")
colons (".")

```
    <li> periods (".") </li>
  </ul>

  <p>referring to the first group of examples in the HTML5 section above, only one is valid: </p>

  <p>
    (<b>div id="sampletext"</b>) ... <b>(/div)</b>  <br>
    (<b>div id="sample-text"</b>) ... <b>(/div)</b>  <br>
    (<b>div id="sample_text"</b>) ... <b>(/div)</b>  <br>
    (<b>div id="sample:text"</b>) ... <b>(/div)</b>  <br>
    (<b>div id="sample.text"</b>) ... <b>(/div)</b>  <br>
  </p>

  <p><b>Then Output Is:</b></p>

  <!-- emmet code is did[id="Sampletext"]*5{...}-->

  <did id="sampletext"> ... </did>  <br>
  <did id="sample-text"> ... </did>  <br>
  <did id="sample_text"> ... </did>  <br>
  <did id="sample:text"> ... </did>  <br>
  <did id="sample.text"> ... </did>  <br>

  <p> Again, if it doesn't start with a letter (uppercase or lowercase), it's not valid. </p>

  <h4> For a Class </h4>

  <p> The rules for classes are essentially the same as for an id. The difference is that class values <i>do not</i> need to be unique in the document. </p>
```

 Referring to the examples above, although this is not valid in the same document:

(div id="results") ... (/div)
(div id="results") ... (/div)

Then Output Is:

...
...

 This is perfectly okay:

(div id="results") ... (/div)
(div id="results") ... (/div)

 Then Outpun is:

...
...

Importat Note: How ID and Class values are treated outside of HTML

 Keep in mind that the rules and examples above apply within the context of HTML

 Using numbers, punctuation or special characters in the values of an id or a class may cause trouble in the other contexts, such as CSS, JavaScript and regular expressions.

 For examples, although the following id is valid in HTML5:

<!-- Emmet HTML code is div[id="9lions"]{...}-->

...

 ... it is invalid in CSS:

 <u>4.1.3 Characters and case</u>

 In CSS, identifiers (including element names, classes, IDs in selectors) can contain only the characters [a-zA-Z0-9] and ISO 10646 characters U+OOAQ and higher, plus the hypen (-) and the underscore(_); they cannot start with a digits, two hypens, or a hypen followed by a digit (emphasis added)

 In Most cases you may be able to escape characters in contexts where they have restrictions o special meaning.

<h4> W3C References </h4>

<!-- Emmet HTML code
is ul>li{<u>3.2.5.1Theidattribute</u>}+li{<u>3.2.5.7Theclassattribute</u>}+li{<u>6.2 SGMLBasictypes</u>} -->

<u>3.2.5.1The id attribute</u>

<u>3.2.5.7 The class attribute</u>

<u>6.2 SGML Basic types</u>

 <!-- Spacing line like enter -->

```
<!-- Chapter 8.4 - Problem related to duplicated IDs -->
<h3> 8.4 - Problem related to duplicated IDs </h3>

<p>Having more than one element with the same ID is a hard to troubleshoot problem. The HTML parses will usually try to render the page in any case. usually no error occurs. But the pace could end up in a mis-behaving web page. </p>

In this example:

<div id="aDiv">a</div>
<div id="aDiv">b</div>

<p>CSS selectors still work</p>

#aDiv {
   color: red;
}

<p>But JavaScript fails to handle both elements:</p>

<b>var</b> html = document.getElementById("aDiv").innerHTML;

In this case html variable bears only the first div content ("a")

</body>
</html>
```